

Ablation Study: SDCN vs SDCN-AE vs SDCN+SemRefine (SBERT)

Dataset	NMI	NMI	NMI	ARI	ARI	ARI	ACC	ACC	ACC	Purity	Purity	Purity
	SDCN	AE	SemRefine	SDCN	AE	SemRefine	SDCN	AE	SemRefine	SDCN	AE	SemRefine
bbcnews	0.8516	0.7258	0.8172	0.8769	0.7385	0.8013	0.9474	0.8804	0.9092	0.9474	0.8804	0.9092
reuters	0.5055	0.5868	0.4488	0.3372	0.4907	0.2469	0.4565	0.6763	0.4101	0.8047	0.8633	0.7859
20newsgroups	0.1527	0.4487	0.1975	0.0420	0.2180	0.0699	0.1382	0.4210	0.1607	0.1483	0.4304	0.1684
agnews	0.5284	0.3467	0.4458	0.5067	0.2969	0.4137	0.7292	0.6188	0.6345	0.7292	0.6188	0.6345
dbpedia	0.7473	0.6531	0.7714	0.6029	0.5267	0.6580	0.6909	0.6696	0.7529	0.6956	0.6950	0.7610

Green cells = best value among SDCN / AE / SemRefine for that metric and dataset. Encoder: SBERT.